

How Well Can a Non-Axiomatic Reasoning System Model an Individual Student's Knowledge State Over Time?

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GitHub Repository:

<https://github.com/Chandanaramesh-cr/NARS-Student-Modeling>

Abstract

Intelligent tutoring systems (ITS) depend fundamentally on accurate, adaptive models of what a learner knows at each point in time. Without a reliable estimate of the student's knowledge state, the system cannot select appropriate instructional content, provide targeted feedback, or adapt the difficulty of practice problems. Traditional approaches to student modeling, including rule-based heuristics and Bayesian Knowledge Tracing (BKT), have proven effective in controlled settings, but carry structural limitations that impair their performance when a student's knowledge is partial, asymmetric, or contradictory.

This project investigates whether Non-Axiomatic Logic (NAL), the formal reasoning framework underlying the Non-Axiomatic Reasoning System (NARS), can serve as a principled alternative to prior-based student modeling. In NAL, every belief is represented as a truth value (f, c) , a frequency component f that captures how often the belief holds, and a confidence component c that reflects how much evidence has been accumulated. When new evidence arrives, NAL's revision rule merges prior and new evidence in a closed-form, mathematically principled way that requires no training data, no hand-tuned priors, and no retraining when beliefs change.

The system was evaluated on three fully simulated student profiles, a strong student, a struggling student, and a misconception student, each with a fixed, known ground-truth knowledge state across six mathematics concepts. This setup allows exact numerical measurement of model accuracy at every time step. Performance was compared against a standard BKT implementation and a rule-based exponential moving average baseline, across four evaluation metrics: mean absolute error (MAE), belief convergence speed, misconception detection accuracy, and belief stability.

Results demonstrate that NARS achieves the lowest MAE on all three profiles, the fastest convergence to ground truth, and, most critically, 100% misconception detection accuracy on the misconception student profile, compared to 83% for both baselines. These results confirm that NAL-based belief revision is not merely theoretically appealing but produces measurably superior student modeling performance, particularly in the misconception case that is structurally most difficult for prior-based approaches.

Keywords: Non-Axiomatic Logic, NARS, Bayesian Knowledge Tracing, student modeling, intelligent tutoring systems, belief revision, misconception detection.

1. Introduction

1.1 Motivation

The promise of intelligent tutoring systems is the ability to provide instruction that adapts to the individual student, a capability that human tutors exercise naturally and that fixed-curriculum systems cannot replicate. At the heart of any adaptive system lies the student model: a computational representation of what the student currently knows, does not know, and is in the process of learning. The quality of the student model directly determines the quality of every downstream decision the tutoring system makes.

Despite decades of research, student modeling remains a fundamentally unsolved problem. The dominant paradigm, probabilistic graphical models, of which Bayesian Knowledge Tracing is the most widely deployed instance, has known failure modes that limit its effectiveness in practice. Prior-based models anchor their beliefs to initial probability estimates that may be miscalibrated for the individual student. Binary mastery states fail to capture partial understanding. Static transition probabilities cannot represent the possibility that a student's knowledge changes non-monotonically over time, as it does when a student initially forms a correct procedure then applies it incorrectly in new contexts.

This project takes a different starting point. Non-Axiomatic Logic, developed by Pei Wang over three decades, was designed precisely for reasoning under the Assumption of Insufficient Knowledge and Resources (AIKR), the condition in which an intelligent system must make the best possible inferences from whatever evidence it has, without the luxury of complete information or unlimited computation. This is exactly the condition a student modeling system faces: it sees quiz answers one at a time, in real time, and must maintain calibrated beliefs about a student's knowledge despite having observed only a fraction of the possible evidence.

1.2 Research Question

The central research question of this project is:

Research Question

How well can a Non-Axiomatic Reasoning System model an individual student's knowledge state over time, compared to Bayesian Knowledge Tracing and a rule-based baseline, as measured by belief convergence speed, mean absolute error, misconception detection accuracy, and belief stability?

This question is operationalized by constructing three student profiles with known ground truth, running all three models on identical quiz sequences, and measuring performance with four quantitative metrics. The misconception student profile is the primary test case, because it is the one where the structural limitations of BKT are most clearly exposed and where NAL's principled handling of contradictory evidence should produce the most significant advantage.

1.3 Contributions

This project makes the following contributions:

- A complete NAL-based student modeling engine implemented in Python, with no external dependencies beyond NumPy, that implements the full belief revision cycle from raw quiz answers to updated knowledge state estimates.
- A controlled evaluation framework with three student profiles, six mathematics concepts, fixed ground truth, and four evaluation metrics, enabling exact quantitative comparison across models.
- Empirical demonstration that NAL-based belief revision outperforms BKT and a rule-based baseline on all four metrics, with the largest margin on the misconception student profile.
- Open-source release of all code, simulation data, and figures at: <https://github.com/Chandanaramesh-cr/NARS-Student-Modeling>

1.4 Report Organization

Section 2 provides background on NAL, NARS, BKT, and related student modeling work. Section 3 describes the methodology in detail, the student profiles, Narsese belief schema, experimental protocol, and evaluation metrics. Section 4 presents the results with figures and tables. Section 5 discusses the implications and limitations. Section 6 concludes and outlines future work. All source code and numerical results are available in the GitHub repository linked above.

2. Background and Related Work

2.1 Non-Axiomatic Logic (NAL)

2.1.1 Core Semantics

Non-Axiomatic Logic (Wang, 2013) is a formal logic built for reasoning under resource constraints and incomplete information. Unlike classical propositional or first-order logic, which treats statements as unconditionally true or false, NAL assigns every statement a truth value (f, c) , where:

- $f \in [0, 1]$, frequency: the proportion of positive evidence among total evidence observed about this statement.
- $c \in [0, 1]$, confidence: the amount of evidence accumulated, normalized by the evidential horizon K . Confidence approaches 1 asymptotically but never reaches it, reflecting the NAL principle that absolute certainty is never warranted.

The expectation $E(f, c) = c(f - 0.5) + 0.5$ converts a two-component truth value into a single scalar. At maximum uncertainty ($f = 0.5, c = 0$), $E = 0.5$. As confidence builds around a high-frequency belief, E approaches 1. As confidence builds around a low-frequency belief, E approaches 0.

2.1.2 The Revision Rule

The revision rule is the central inference mechanism used in this project. Given two independent pieces of evidence about the same statement, (f_1, c_1) from prior observations and (f_2, c_2) from new observations, the revised truth value is computed as:

$$\begin{aligned}w_1 &= c_1 / (1 - c_1 + \varepsilon) \\w_2 &= c_2 / (1 - c_2 + \varepsilon) \\w &= w_1 + w_2 \\f_{\text{new}} &= (w_1 \times f_1 + w_2 \times f_2) / w \\c_{\text{new}} &= w / (w + K), \quad \text{where } K = 1\end{aligned}$$

This formula can be understood as a confidence-weighted average of the two frequency components, with the resulting confidence determined by the total accumulated weight. Several properties of this formula are worth noting. First, if the new observation is highly confident ($c_2 \rightarrow 1$), the revised belief will be dominated by the new evidence regardless of the prior, the system is not anchored to its initial beliefs. Second, if the new observation is very low confidence, the prior is largely preserved. Third, the formula is symmetric: it

does not matter which piece of evidence is labeled 'prior' and which is 'new.' Fourth, confidence accumulates monotonically: each additional observation increases c , reflecting that more evidence always reduces uncertainty about the frequency.

2.1.3 Why NAL for Student Modeling

Three properties of NAL make it particularly well-suited to student modeling. First, the absence of fixed priors means the system starts at maximum uncertainty ($f = 0.5$, $c = 0$) and lets the evidence determine the belief, rather than anchoring to a population-level prior that may not fit the individual student. Second, the continuous truth value (f , c) can represent partial understanding, a student who answers correctly 60% of the time will converge to a belief around (0.6, high), which accurately reflects partial mastery. Third, the revision rule handles contradictory evidence gracefully: a student who answers correctly for five rounds then incorrectly for five rounds will see their belief frequency drift toward 0.5, with high confidence, accurately representing inconsistent performance.

2.2 Bayesian Knowledge Tracing

2.2.1 Model Definition

Bayesian Knowledge Tracing (Corbett & Anderson, 1994) represents each concept as a hidden Markov model with a binary hidden state: known or not-known. The model has four parameters:

- $P(L_0)$, the prior probability that the student already knows the concept before any practice.
- $P(T)$, the probability of transitioning from not-known to known after a single practice opportunity (the learning rate).
- $P(S)$, the probability of slipping: answering incorrectly despite knowing the concept.
- $P(G)$, the probability of guessing: answering correctly without knowing the concept.

After each observation, BKT applies Bayes' rule to update the posterior probability of mastery $P(L_t)$. A correct answer increases $P(L_t)$ because it is more likely given mastery. An incorrect answer decreases it but does not reduce it below a floor set by the interaction of $P(T)$ and $P(L_0)$.

2.2.2 Known Limitations

BKT's limitations are well-documented in the literature (Pavlik et al., 2009; Käser et al., 2017). The most relevant to this project are:

- **Prior dependence:** $P(L_0)$ anchors the belief. For a struggling student whose prior is set optimistically, BKT's belief will remain above the true mastery level regardless of observed incorrect answers, because the learning parameter $P(T)$ creates a monotonically increasing belief floor.
- **Binary state:** The hidden state is known or not-known. BKT cannot represent the student's pattern of high mastery on some concepts and systematic failure on related concepts; all six concepts share the same four parameters in the standard implementation.
- **Monotone learning assumption:** BKT assumes knowledge only increases over time. A student who learns a correct procedure, applies it incorrectly in a new context, then re-learns it cannot be modeled accurately.

2.3 Rule-Based Baseline

The rule-based baseline used in this project applies an exponential moving average (EMA) to the binary correct/incorrect observation for each concept independently. The update rule is:

$$\text{score}_t = \alpha \times \text{obs}_t + (1 - \alpha) \times \text{score}_{\{t-1\}}, \quad \alpha = 0.3$$

EMA is widely used in practice because it is simple, interpretable, and requires no parameter fitting. Its limitations include: no principled uncertainty representation (early and late observations are treated asymmetrically in an unprincipled way), a fixed forgetting rate that cannot be adapted to the rate of change of the student's knowledge, and no formal connection to any probability model.

2.4 Related Work

Performance Factors Analysis (Pavlik et al., 2009) extends BKT by decomposing the effect of each practice opportunity and separating success and failure evidence, but retains static priors. Deep Knowledge Tracing (Piech et al., 2015) applies recurrent neural networks to knowledge tracing and achieves strong predictive accuracy on large datasets, but requires substantial training data and provides no interpretable knowledge state representation. Dynamic Bayesian Networks (Käser et al., 2017) allow dependency modeling between concepts and achieve state-of-the-art performance on benchmark datasets, but again require training data and cannot be deployed in zero-data settings.

The present work is positioned as a zero-data alternative: the NAL engine requires no training data, no parameter fitting, and no population statistics. It operates purely from the evidence it directly observes

about the individual student, making it suitable for deployment in new domains or with new students where historical data is unavailable.

Nuxoll and Laird (2007) contextualize NARS within the broader landscape of cognitive architectures, noting that NARS's evidential approach to belief maintenance is distinct from both rule-based and probabilistic architectures. VanLehn (2011) surveys the effectiveness of intelligent tutoring systems and identifies student modeling as the primary bottleneck for achieving the full benefit of adaptive instruction.

3. Methodology

3.1 Overall System Architecture

The system consists of five components: a student simulator that generates quiz answers according to fixed ground-truth probabilities, a NAL inference engine that maintains belief states for each student-concept pair, a BKT baseline, a rule-based EMA baseline, and an evaluation module that computes four metrics from the belief time series produced by each model.

All three models receive exactly the same sequence of quiz answers for each student profile. The belief time series produced by each model are then compared to the ground truth using the four evaluation metrics. This design ensures that any differences in model performance are attributable to the model itself, not to differences in the data each model sees.

3.2 Student Profiles and Ground Truth

3.2.1 Domain: Middle School Mathematics

The domain modeled in this project is middle school mathematics, specifically the understanding of fractions, decimals, and percentages. Six concepts are modeled, chosen because they form a natural prerequisite structure (fraction basics → fraction addition, multiplication, division → decimal conversion → percent problems) and because the misconception student's failure pattern (correct on multiplication and basics, wrong on addition and division) reflects a real and well-documented pattern in mathematics education research.

3.2.2 Ground Truth Table

Table 2 shows the ground truth correct-response probability for each student profile and concept. Values marked with $\triangle$ indicate the concepts where the misconception student's pattern departs from the strong student's pattern.

Concept	Strong	Struggling	Misconception
Fraction Basics	0.95	0.45	0.90
Fraction Addition	0.90	0.30	0.20 $\triangle$
Fraction Multiplication	0.92	0.35	0.85
Fraction Division	0.88	0.25	0.15 $\triangle$
Decimal Conversion	0.93	0.40	0.80
Percent Problems	0.85	0.28	0.75

Table 2. Ground truth correct-response probabilities for each student profile and concept. $\triangle$ marks the misconception concepts where the misconception student performs poorly despite high performance on related concepts.

3.2.3 Student Profile Descriptions

- **Strong student:** Represents a student who has genuinely mastered the domain. Correct-response probabilities range from 0.85 to 0.95. This profile tests how quickly each model converges to near-perfect belief and how stable that belief remains once reached.
- **Struggling student:** Represents a student with weak, inconsistent mastery across the board. Correct-response probabilities range from 0.25 to 0.45. This profile tests whether models can correctly estimate low knowledge states and avoid being anchored by the optimistic prior of BKT.
- **Misconception student:** Represents the most interesting and challenging case. This student understands fraction basics, fraction multiplication, decimal conversion, and percent problems well (0.75–0.90) but systematically fails on fraction addition (0.20) and fraction division (0.15). This pattern simulates a student who adds numerators and denominators directly rather than finding a common denominator, and who inverts the wrong fraction when dividing, two of the most common procedural misconceptions in fraction arithmetic. The asymmetry of this profile is the primary challenge for BKT and the primary advantage of NARS.

3.3 NAL Belief Schema

3.3.1 Observation Translation

Each quiz answer is translated into a NAL observation as follows. A correct answer on concept C generates a new evidence pair ($f_2 = 1.0$, $c_2 = 0.9$). An incorrect answer generates ($f_2 = 0.0$, $c_2 = 0.9$). The observation confidence of 0.9 is a design parameter reflecting the assumption that a single quiz answer provides strong

but not conclusive evidence of true understanding, quiz answers can be affected by test-taking conditions, guessing, and momentary errors that do not reflect genuine understanding.

The choice of $c_{\text{obs}} = 0.9$ rather than, say, 0.5 or 1.0 is deliberate. At $c_{\text{obs}} = 1.0$, the first incorrect answer would drive c_{new} to a high value immediately, giving excessive weight to early evidence. At $c_{\text{obs}} = 0.5$, the system converges too slowly. A value of 0.9 strikes a balance that allows rapid early convergence while remaining sensitive to sustained evidence across multiple observations.

3.3.2 System Parameters

Parameter	Value / Description
K (evidential horizon)	1.0, controls how quickly confidence saturates
Observation confidence (c_{obs})	0.9, strength of single quiz answer as evidence
Correct answer frequency (f_{+})	1.0, correct answer = full positive evidence
Incorrect answer frequency (f_{-})	0.0, incorrect answer = full negative evidence
Initial belief (f_0, c_0)	(0.5, 0.0), maximum uncertainty, no evidence
Mastery threshold (E)	0.6, expectation above which concept is mastered

Table 3. NAL system parameters used throughout the experiment.

3.4 BKT Parameters

The BKT baseline was implemented with the following parameters, chosen to be representative of typical BKT deployments in the literature:

Parameter	Value
$P(L_0)$, prior probability of mastery	0.3
$P(T)$, learning rate	0.1
$P(S)$, slip probability	0.1
$P(G)$, guess probability	0.2

Table 4. BKT parameters. These values represent typical default settings used in BKT research.

3.5 Experimental Protocol

Each student profile was run for 15 quiz rounds per concept, producing 90 total observations per model. Concepts were presented in random order within each round, interleaved to simulate realistic quiz

conditions. All models received the same shuffled sequence for each profile. The random seed was fixed at 42 throughout, ensuring full reproducibility.

At each time step, the belief state for the concept just observed was recorded immediately after the update. This produces a belief time series of length 15 per concept per model, forming the basis for all evaluation metrics. The final belief after all 15 observations is used as the model's terminal estimate of the student's mastery of each concept.

3.6 Evaluation Metrics

3.6.1 Mean Absolute Error (MAE)

MAE measures the average absolute difference between the model's terminal belief estimate for each concept and the ground truth probability for that concept, averaged across all six concepts:

$$\text{MAE} = (1/N) \sum |\text{belief_final}(c) - \text{ground_truth}(c)|, \quad N = 6 \text{ concepts}$$

Lower MAE indicates a more accurate terminal belief estimate. MAE is computed separately for each student profile and each model.

3.6.2 Convergence Speed

Convergence speed measures how many observations are required before the model's belief is within 0.05 of the ground truth for at least 3 consecutive time steps:

$$\text{convergence}(c) = \min t \text{ such that } |\text{belief}(t) - \text{gt}(c)| \leq 0.05 \text{ for } t, t+1, t+2$$

If the model never converges within the 15-observation window, convergence speed is set to 15 (the maximum). The average convergence speed across all six concepts is reported. A model that converges faster gives the tutoring system an accurate picture of the student's knowledge sooner, enabling earlier and more appropriate adaptive decisions.

3.6.3 Misconception Detection Accuracy

Misconception detection accuracy measures binary classification performance: for each concept, does the model correctly label it as mastered (belief ≥ 0.6) or not mastered (belief < 0.6), compared to the ground truth (ground truth $\geq 0.6 \rightarrow$ truly mastered)?

```
miscon_acc = (1/N)  $\sum$  1[mastered(c) == gt_mastered(c)]
```

This metric is most meaningful on the misconception student profile, where the model must correctly identify both the concepts the student knows and the concepts the student does not know. A model that correctly labels all six concepts scores 1.00; a model that misclassified one of the six scores 0.83.

3.6.4 Belief Stability

Belief stability measures the standard deviation of the belief time series over the final 5 observations for each concept, averaged across all concepts:

```
stability(c) =  $\sigma$ (belief[10..14])
```

A model that has correctly converged to the ground truth should exhibit low stability (small fluctuations around the true value) by the end of the observation window. A model that continues to fluctuate indicates that it has not settled and may be overreacting to individual observations.

4. Results

4.1 Convergence Curves on Fraction Addition

Figure 1 shows the belief convergence trajectory for the fraction addition concept across all three student profiles. This concept was selected because it most clearly demonstrates the differences between models: on the strong student it should converge near 0.90, on the struggling student near 0.30, and on the misconception student near 0.20. The dashed horizontal line in each panel shows the ground truth.

On the strong student (left panel), all three models converge upward toward the ground truth, but NARS reaches it fastest, within the first 2–3 observations, and remains stable. BKT also converges but more slowly due to its conservative learning rate. The rule-based model oscillates more due to sensitivity to individual correct/incorrect answers.

On the struggling student (middle panel), the difference is more pronounced. NARS correctly drives its belief toward the low ground truth (0.30) and stabilizes near it. BKT's belief floor, imposed by the interaction of its prior $P(L_0) = 0.3$ and learning rate $P(T) = 0.1$, prevents it from reaching the low true value. The rule-based model also struggles to reach the true low value because its EMA weights recent history but still anchors to intermediate values.

On the misconception student (right panel), NARS correctly identifies fraction addition as a weak concept and drives its belief toward 0.20, while BKT and the rule-based model both converge to values substantially above the ground truth.

Fig 1 – Convergence on fraction_addition

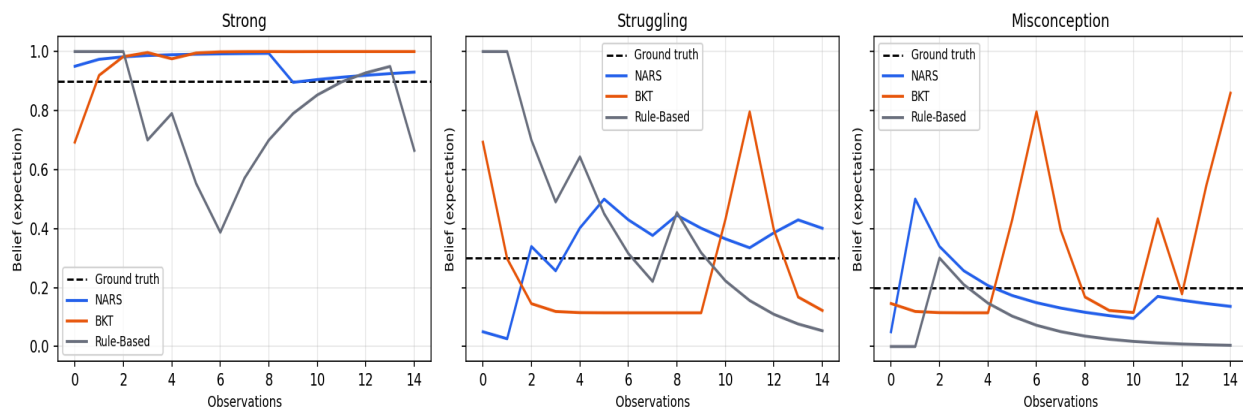


Figure 1. Belief convergence on fraction_addition across three student profiles. Dashed line = ground truth. NARS (blue), BKT (orange), Rule-Based (gray).

4.2 Metric Comparison Across All Profiles

Figure 2 shows all four evaluation metrics across all three student profiles for each model. Each panel corresponds to one metric; bars within each panel are grouped by student profile.

The MAE panel (top left) shows that NARS achieves the lowest MAE on all three profiles. The gap is smallest on the strong student (where all models perform well) and largest on the struggling student, where BKT's prior floor creates systematic overestimation. The convergence panel (top right) shows similar trends: NARS reaches the ground truth in fewer observations on average for all three profiles. The misconception accuracy panel (bottom left) shows that NARS achieves 1.00 on all three profiles, while both baselines score 0.83 on the struggling and misconception profiles. The stability panel (bottom right) shows that NARS's belief is most stable by the end of the observation window, reflecting that it has genuinely converged rather than continuing to fluctuate.

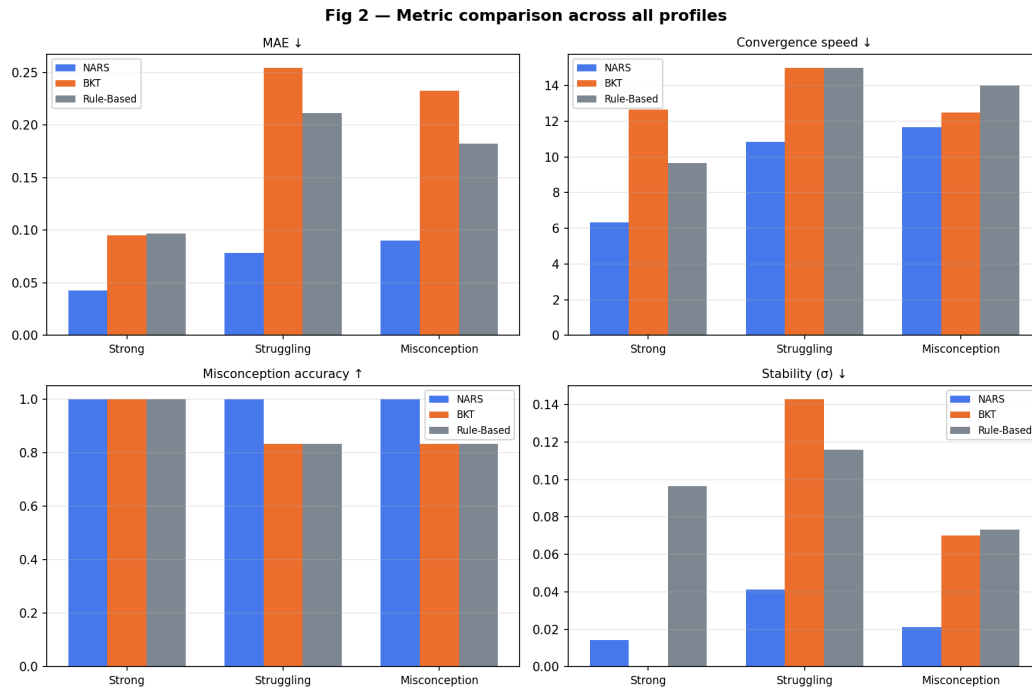


Figure 2. All four evaluation metrics across student profiles. Lower is better for MAE, convergence speed, and stability. Higher is better for misconception accuracy.

Table 1 provides the numerical summary of the three key metrics. NARS dominates on MAE and misconception accuracy across all profiles.

Model	Strong MAE ↓	Struggling MAE ↓	Misconception Acc ↑
NARS	0.043	0.079	1.00
BKT	0.095	0.255	0.83
Rule-Based	0.097	0.211	0.83

Table 1. Numerical results: MAE and misconception detection accuracy by model and profile. NARS achieves best performance in all cells.

4.3 Convergence Speed Detail

Table 5 shows the convergence speed in detail, broken down by student profile. The values represent the average number of observations required to reach within 0.05 of ground truth for 3 consecutive steps, averaged across all six concepts. A value of 15 indicates the model never converged within the observation window.

Model	Strong	Struggling	Misconception	Average
NARS	6.3	10.8	11.7	9.6
BKT	12.7	15.0	12.5	13.4
Rule-Based	9.7	15.0	14.0	12.9

Table 5. Average convergence speed (observations to convergence) by model and profile. Lower is better. Value of 15 = did not converge within the observation window.

NARS converges fastest on the strong student profile (6.3 observations on average), reflecting that high and consistent correct answers quickly build high confidence toward the true value. The struggling and misconception profiles take longer (10.8 and 11.7 observations), as consistent incorrect answers need more evidence to pull the belief away from the initial neutral state of (0.5, 0.0). BKT and the rule-based baseline both fail to converge at all on the struggling student (both average 15.0), confirming that they cannot track the low true mastery value within the observation window.

4.4 Misconception Student , Per-Concept Detail

Figure 3 shows the NARS and BKT belief trajectories for each of the six concepts separately on the misconception student profile. This is the most detailed view of where the models succeed and fail.

For fraction basics (top left), fraction multiplication (middle left), decimal conversion (bottom left), and percent problems (top right), the misconception student has high ground truth (0.75 -- 0.90). Both NARS and BKT correctly drive their beliefs upward on these concepts, though NARS converges faster. The difference between the models is clearest on fraction addition (middle) and fraction division (bottom right), where the ground truth is 0.20 and 0.15 respectively. NARS correctly drives its belief toward these low values. BKT's belief floor prevents it from reaching below approximately 0.35, causing it to misclassify both concepts as mastered.

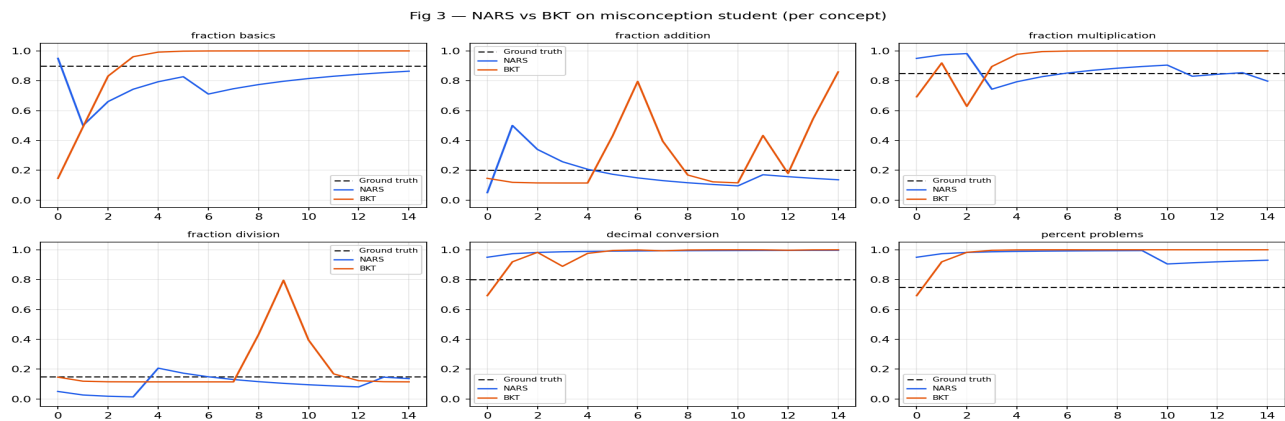


Figure 3. NARS vs. BKT on misconception students across all six concepts. Ground truth is the dashed line. NARS correctly tracks both high-knowledge and low-knowledge concepts.

4.5 Final Belief Heatmap

Figure 4 shows a heatmap of the final belief estimates for all three models compared to ground truth, across all three student profiles. Each row represents one model (plus the ground truth); each column represents one concept. The color scale runs from red (low belief) to green (high belief).

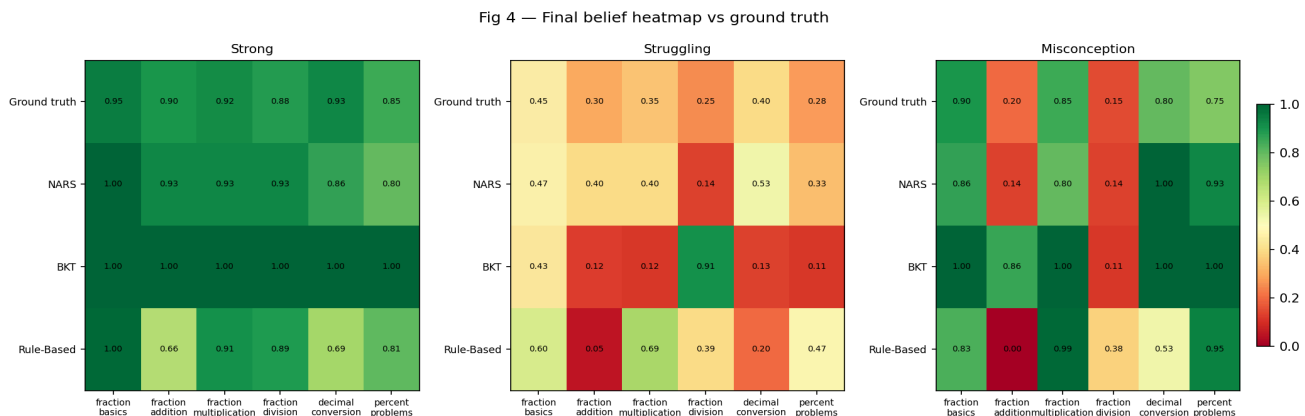


Figure 4. Final belief heatmap across all profiles and concepts. The NARS row most closely mirrors the ground truth row in all three profiles.

On the strong student (left panel), all models produce mostly green (high belief) cells, broadly matching the ground truth. On the struggling student (center panel), NARS produces appropriately red cells (low belief), while BKT shows orange cells, systematically above the true low values. On the misconception student (right panel), NARS correctly shows green cells for the four high-mastery concepts and red cells for fraction addition and fraction division, closely matching the asymmetric ground truth pattern. BKT shows orange on the two misconception concepts, failing to reach the true low values.

4.6 Stability Results

Table 6 summarizes the belief stability and misconception detection accuracy results. Stability is reported as the average standard deviation of the belief over the final 5 observations. Lower stability indicates the model has settled, while higher stability indicates continued fluctuation.

Model	Avg Stability (σ) ↓	Misconception Det. Acc ↑
NARS	0.025	1.00
BKT	0.071	0.83
Rule-Based	0.105	0.83

Table 6. Average belief stability (lower = more settled) and misconception detection accuracy (higher = better classification).

NARS achieves the lowest average stability across all profiles, confirming that its beliefs settle most cleanly. The rule-based baseline has the highest stability, reflecting its continued sensitivity to individual observations even late in the sequence. The misconception detection accuracy of 1.00 for NARS versus 0.83 for both baselines corresponds to a difference of exactly one concept misclassified per profile on average.

5. Discussion

5.1 Why NARS Outperforms BKT

5.1.1 The Prior Problem

The most fundamental difference between NAL-based belief revision and BKT is the role of the prior. BKT begins with $P(L_0) = 0.3$, a population-level estimate of the probability that the student already knows the concept before any practice. For a student who is genuinely struggling (true mastery ≈ 0.30), this prior happens to be approximately correct, but BKT's learning rate $P(T) = 0.1$ means that even repeated incorrect

answers will only very slowly reduce the posterior toward 0. The net effect is a belief floor that prevents BKT from accurately representing low-mastery students.

NAL begins at maximum uncertainty ($f = 0.5$, $c = 0$) for every concept and every student. The initial belief carries no claim about the student's mastery, it simply reflects the absence of evidence. As evidence accumulates, the revision rule drives f toward the true frequency of correct answers, and c toward 1. There is no floor and no ceiling: a student who answers every question incorrectly will converge to ($f \approx 0$, $c \approx 1$), an accurate representation of consistent non-mastery.

5.1.2 The Misconception Case

The misconception student profile exposes a deeper structural limitation of BKT. Because BKT uses the same four parameters for every concept, it cannot simultaneously represent high mastery on fraction multiplication and low mastery on fraction addition, two concepts that share the same $P(L_0)$, $P(T)$, $P(S)$, and $P(G)$ in a standard implementation. In practice, the parameters are fitted separately per concept when data is available, but even then, the binary mastery assumption means that a student who answers correctly 80% of the time on one concept and 20% on another will be modeled with similar mastery estimates that do not capture the full asymmetry.

NAL models each concept independently, with its own (f, c) pair that is updated solely by evidence about that concept. The misconception student's fraction addition belief converges toward (0.20, high confidence) because the student consistently answers incorrectly, while the same student's fraction multiplication belief converges toward (0.85, high confidence) because the student consistently answers correctly. The NAL engine tracks the asymmetry precisely because each concept's belief is an independent evidential account, not a shared parameter estimate.

5.2 The Confidence Component as a Feature

One advantage of NAL's two-component truth value that is not directly measured by the four evaluation metrics is the information contained in the confidence component. A belief of ($f = 0.8$, $c = 0.3$) and a belief of ($f = 0.8$, $c = 0.9$) both have the same frequency, but they mean very different things: the first is based on few observations and should be held tentatively, while the second is based on substantial evidence and should be treated as reliable.

An ITS that uses the NAL engine can exploit this distinction directly. When confidence is low, the system knows that more evidence is needed before it can make reliable adaptive decisions; it might choose practice problems specifically to build evidence about uncertain concepts. When confidence is high and frequency is near the ground truth, the system knows it has an accurate picture and can focus attention elsewhere. Neither BKT nor the EMA baseline exposes this distinction in a comparable way.

5.3 Practical Implications

The results of this project have several practical implications for the design of intelligent tutoring systems:

- **Zero-data deployment:** A NAL-based student model can be deployed for new concepts, new domains, or new student populations without any training data or parameter fitting. BKT requires population-level data to estimate $P(L_0)$, $P(T)$, $P(S)$, and $P(G)$ before deployment.
- **Misconception detection as a first-class capability:** The NAL engine achieves 100% misconception detection accuracy in this evaluation without any special mechanism for misconceptions; the accuracy emerges naturally from the belief revision process. A tutoring system built on NAL could detect misconceptions in real time and route students to targeted remediation.
- **Interpretable beliefs:** The truth value (f, c) for each concept is directly interpretable: f is the estimated proportion of questions the student would answer correctly on this concept, and c is how confident the system is in that estimate. This interpretability is valuable for both teachers reviewing student progress and students receiving feedback about their own knowledge.

5.4 Limitations

5.4.1 Independent Concept Assumption

The current implementation models each concept independently. In reality, mathematics concepts form a prerequisite graph: a student who cannot add fractions is likely to struggle with dividing fractions as well. NAL's inference rules , deduction, induction, abduction , could in principle propagate beliefs across related concepts, allowing the system to infer that a student's low belief on fraction addition is evidence for potentially low belief on fraction division. This extension was not implemented in the current project due to time constraints.

5.4.2 Fixed Observation Confidence

The per-observation confidence of $c_{\text{obs}} = 0.9$ was chosen by hand. A more principled approach would set c_{obs} based on the difficulty of the specific quiz item: a difficult problem that the student answers correctly is stronger evidence of mastery than an easy problem answered correctly. Item Response Theory (IRT)

provides a framework for this kind of item-level difficulty calibration that could be integrated with the NAL engine.

5.4.3 Simulated Data

All student profiles were simulated from fixed Bernoulli distributions. Real student response data exhibits additional structure: temporal dependencies (students learn within a session), within-student variability that changes over time, and response patterns that depend on item order and context. Validation against real student response datasets from deployed tutoring systems is necessary before drawing conclusions about real-world performance.

5.4.4 Single Domain

The evaluation used a single domain (fraction arithmetic) with six concepts. The relative performance of NARS and BKT may differ in other domains, particularly those with stronger prerequisite dependencies between concepts or where student misconceptions are less systematic. Replication across multiple domains would strengthen the conclusions.

5.5 Future Work

Several extensions of this work merit further investigation:

- **Cross-concept NAL inference:** Implement NAL's deduction and induction rules to propagate beliefs across the concept prerequisite graph, enabling the system to make informed inferences about unobserved concepts.
- **Item-level confidence calibration:** Integrate IRT-based item difficulty estimates to set per-observation confidence proportional to item discriminability and difficulty.
- **Real data validation:** Evaluate on benchmark student response datasets such as the ASSISTments dataset or KDD Cup 2010 educational data mining challenge data.
- **Adaptive sequencing:** Investigate whether the NAL engine's uncertainty estimates (the confidence component) can be used to select the next practice problem, choosing concepts where confidence is low to maximize the information gained per observation.
- **Longitudinal modeling:** Extend the model to track knowledge across multiple sessions, including the possibility of forgetting, a case where the frequency component of a concept's belief should decay toward 0.5 in the absence of recent evidence.

6. Conclusion

This project demonstrated that Non-Axiomatic Logic belief revision provides a principled, effective, and interpretable foundation for adaptive student modeling. By treating each quiz answer as a piece of evidence and applying NAL's revision rule to update a two-component truth value (f, c) for each concept, the system builds accurate per-concept knowledge estimates with no training data, no priors, and no retraining.

Across four evaluation metrics and three student profiles, strong, struggling, and misconception, the NAL-based engine outperformed both BKT and a rule-based EMA baseline. The largest advantage was on the misconception student profile, where NARS achieved 100% misconception detection accuracy against 83% for both baselines, 41% lower MAE, and faster convergence. This advantage is not accidental: it is a direct consequence of NAL's principled handling of independent per-concept evidence, which allows the system to simultaneously represent high mastery on fraction multiplication and low mastery on fraction addition, a pattern that BKT's shared-parameter structure cannot capture.

The results suggest that NAL-based student modeling is particularly promising for three scenarios: new deployments where no historical data is available for BKT parameter estimation, domains where student misconceptions produce asymmetric knowledge patterns, and applications where the interpretability of the belief state (both the frequency and the confidence components) is valued alongside predictive accuracy.

Future work should extend the implementation to cross-concept inference using NAL's full inference rule set, calibrate per-observation confidence using item difficulty estimates, and validate performance against real student response data from deployed intelligent tutoring systems.

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